

WASP-10b

R-1029

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-10_161030 · created 2026-07-30T06:05:58+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2457691.81346171 +/- 5.8e-07 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.2117452 +/- 3.2e-06
Transit depth (Rp/Rs)^2	4.48 +/- 0.00013 [%]
Orbital Inclination (inc)	85.9951 +/- 0.0025
Airmass coefficient 1 (a1)	5.52 +/- 1.66
Airmass coefficient 2 (a2)	-1.1324 +/- 0.00075
Scatter in the residuals of the lightcurve fit is	22.63 %
Best Comparison Star	#3 - [604, 142]
Optimal Method	PSF photometry
Transit Duration (day)	0.037814 +/- 2.4e-05

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.2117452 ± 3.2e-06 vs 0.1592 (deviation 32.76σ)
Duration fit vs published	0.037814 d vs 0.0946 d (deviation 4.72σ)
Mid-time O-C	-7.8 min
Depth SNR	132.0
Residual scatter	22.63 %

Light curve

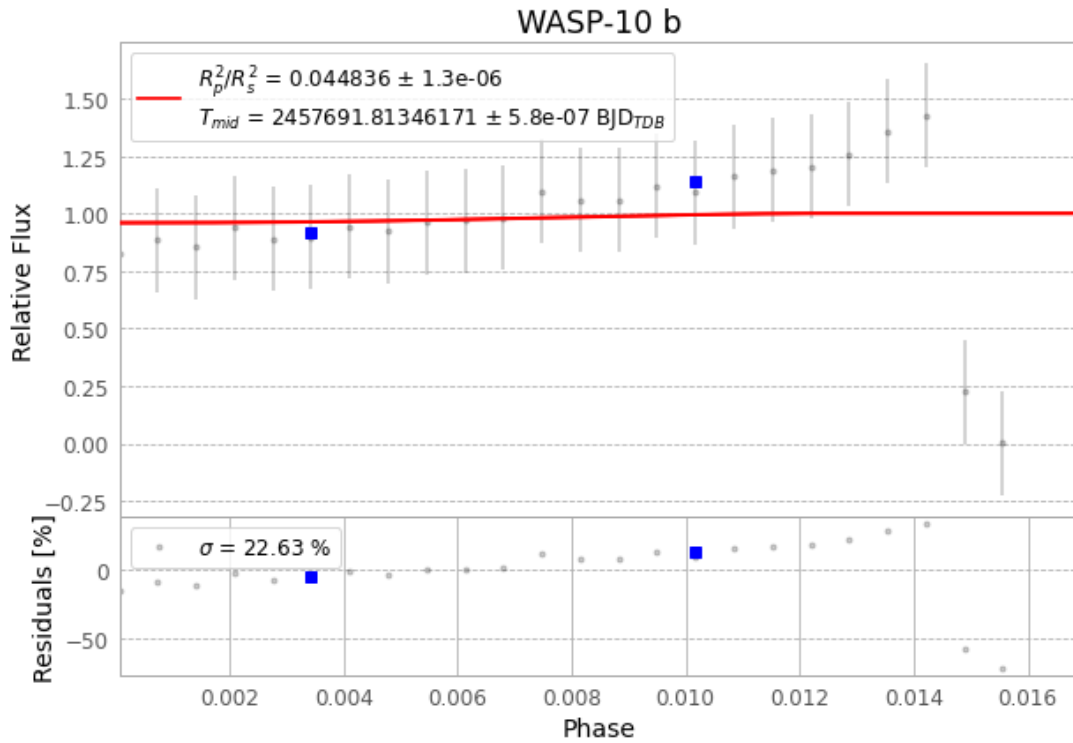


Figure 1 — FinalLightCurve_WASP-10 b_2016-10-30.png (SHA-256 1fad80e804b9c147...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [609, 241], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 8d8ecfc92cede9aa...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit d840746

Data provenance

64 FITS frames (42 MB), WASP-10161030072412.FITS → WASP-10161030103312.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-10-161030.log (84699032045f...), FinalLightCurve_WASP-10 b_2016-10-30.png (1fad80e804b9...), FinalLightCurve_WASP-10 b_2016-10-30.csv (cd204ead4bb0...)

Compute resources

Wall time 125.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-30 06:05Z.